

Contents

Abstract	1
Table of Contents	4
Figure Index	5
Table Index	6
1.1 Aim of the Project	8
1.2 Motivation	8
1.3 Technology Overview	9
1.4 Organization of the Report	10
2.1 State-wise Crop Residue Generation	13
2.2 Limitations of Existing Solutions	13
3.1 Agricultural Waste Management in India	16
3.2 Digital Platforms for Waste Management	16
3.3 Firebase and Real-Time Database Systems	16
3.4 Next.js and Modern Web Frameworks	16
3.5 Carbon Credit Systems	16
3.6 Payment Gateway Integration	17
3.7 IoT and Sensor-Based Waste Monitoring	17
3.8 Mobile-First Web Applications for Rural India	17
3.9 Tailwind CSS and Utility-First Design Systems	17
3.10 Comparative Analysis of Existing Agricultural Platforms	18
3.11 Summary of Literature Review	18
4.1 Project Objectives	20
4.2 Scope of the Project	20
5.1 Hardware Requirements	23
5.2 Software Requirements	23
5.3 Functional Requirements	23
5.4 Non-Functional Requirements	25
6.1 Development Approach	28
6.2 Sprint Breakdown	28
6.3 Architecture Pattern	28
6.4 Development Tools and Environment	29
6.5 Data Flow	29
7.1 Gantt Chart	32
7.2 DFD (Data Flow Diagram)	32
7.3 Flow Diagram	33